Campylobacter transfer from naturally contaminated chicken thighs to cutting boards is inversely related to initial load.

Foods prepared in the kitchen can become cross-contaminated with Campylobacter by contacting raw products, particularly skinned poultry. We measured the percent transfer rate from naturally contaminated poultry legs purchased in supermarkets. Transfer of Campylobacter from skin (n = 43) and from meat (n = 12) to high-density polyethylene cutting board surfaces was quantitatively assessed after contact times of 1 and 10 min. The percent transfer rate was defined as the ratio between the number of Campylobacter cells counted on the cutting board surface and the initial numbers of Campylobacter naturally present on the skin (i.e., the sum of Campylobacter cells on the skin and board). Qualitative transfer occurred in 60.5% (95% confidence interval, 45.5 to 75.4) of the naturally contaminated legs studied and reached 80.6% (95% confidence interval, 63.0 to 98.2) in the subpopulation of legs that were in contact with the surface for 10 min. The percent transfer rate varied from 5 x 10(-2)% to 35.7% and was observed as being significantly different (Kruskall-Wallis test, $P < 0.025$) and inversely related to the initial counts on poultry skin. This study provides quantitative data describing the evolution of the proportion of Campylobacter organisms transferred from naturally contaminated poultry under kitchen conditions. We emphasize the linear relationship between the initial load of Campylobacter on the skin and the value of the percent transfer rate. This work confirms the need for modeling transfer as a function of initial load of Campylobacter on leg skin, the weight of poultry pieces, and the duration of contact between the skin and surface.